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GPT-6-Astra Lights Up Embodied Navigation

Evaluation in Zero-Shot Vision-and-Language Navigation in Continuous Environments

Guangzhao Dai[1], Qianru Sun[1], Qi Wu[2], Bin Zhu[1†]
[1]School of Computing and Information Systems, Singapore Management University
[2]Australia Institute for Machine Learning
[†]Corresponding author and project lead

Website: https://daiguangzhao.github.io/gpt-6-astra-for-vln/

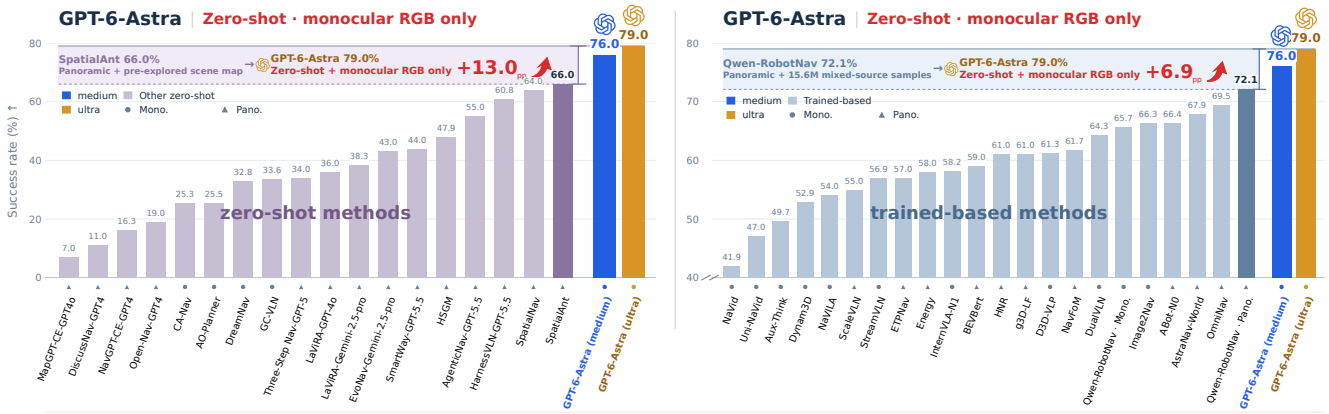

Figure 1. **Strong zero-shot navigation performance of GPT-6-Astra.** With monocular RGB, SR is 76.0% (medium) and 79.0% (ultra). Ultra exceeds the best reported zero-shot and trained results by 13.0 and 6.9 percentage points. Following the common-used zero-shot evaluation protocol [10, 24, 25], we use the 100-episode R2R-CE subset [24] for evaluation. The trained-based results use the full validation-unseen split. Although comparsion not strictly fair, these comparisons help the VLN community assess the competitiveness and research value of zero-shot navigation with advanced foundation models.

Abstract

We investigate whether GPT-6-Astra, a general-purpose foundation model, can navigate unfamiliar environments using its own perception, reasoning, and decision-making capabilities. Our evaluation focues on zero-shot vision-and-language navigation in continuous environments (VLN-CE) through a minimal interface in the Codex harness, aiming to unleash GPT-6-Astra's full potential for navigation. Using monocular RGB, GPT-6-Astra decides when to observe, how to move, and when to stop, without navigation-specific fine-tuning, a trained waypoint predictor, or a pre-built scene map. Our evaluation yields four main findings. First, ***GPT-6-Astra achieves strong zero-shot navigation performance using only monocular RGB observations****. On the common-adopted zero-shot R2R-CE benchmark, ultra reasoning achieves a success rate of* ***79.0%****, exceeding the strongest reported zero-shot and supervised success rates by* ***13.0*** *and* ***6.9*** *percentage points, respectively. Second,* ***GPT-6-Astra advances multi-stage language instructions into coherent, adaptive navigation*** *by grounding spatial relations, tracking task progress, and revising its actions. Third,* ***reliable route execution and goal verification remain challenging, even with ultra reasoning****. Plausible local landmark matches do not consistently lead to correct task completion. Fourth,* ***these capabilities motivate rethinking the role of embodied learning****. Future VLN research should build on foundation models to advance generalizable and reliable embodied intelligence.*

1. Introduction

Can a general-purpose foundation model navigate an unfamiliar environment by directing its own interaction with it? Here, we use GPT-6-Astra to study this question through Vision-and-Language Navigation (VLN), where an agent follows natural-language instructions to reach a destination in a visual environment [3]. Completing this task requires the agent to follow an language instruction to recognize landmarks, ground spatial relations, track its progress, and choose actions. In continuous environments, it must also coordinate turns and forward movements, inspect uncertain situations, recover from mistakes, and stop at the appropri-

ate location [17]. VLN therefore tests whether a foundation model can bring perception, reasoning, memory, and action together over an extended interaction.

Early VLN research developed navigation policies trained on annotated instructions and trajectories [3, 12]. With the emergence of models such as GPT-4, Gemini-2.5-Pro, and Qwen2.5-VL, attention increasingly shifted toward transferring general language and visual capabilities to navigation [4, 11]. Many approaches adapt these capabilities through navigation-specific training or human-designed workflows for planning, memory, and action selection [24, 25, 36, 39]. This report does not introduce a new navigation method. Following the harness-based setup of [47], we use codex harness[1] to run GPT-6-Astra with a minimal interface for monocular RGB observations and primitive actions. Here, codex only sustains the session, while GPT-6-Astra makes the navigation decisions. This setting allows us to examine the model's capabilities as expressed through its own interaction with the environment. Our evaluation yields four main findings.

(1) GPT-6-Astra reaches new heights in VLN performance using only monocular RGB observations in a zero-shot way. Following the evaluation protocol of prior zero-shot works [10, 24–26], we evaluate GPT-6-Astra on the zero-shot R2R-CE benchmark[2]. GPT-6-Astra (medium reasoning) and GPT-6-Astra (ultra reasoning) achieve success rates of **76.0%** and **79.0%**, with corresponding SPL scores of **67.9%** and **69.5%**. As shown in Figure 1, GPT-6-Astra (ultra reasoning) surpasses the reported success rates of SpatialAnt and Qwen-RobotNav by **13.0** and **6.9** percentage points, respectively [43, 44]. Although evaluation subsets and system configurations differ across these methods, the comparison highlights strong performance using monocular RGB alone, without navigation-specific fine-tuning, a trained waypoint predictor, or a pre-built scene map.

(2) GPT-6-Astra can translate multi-stage language instructions into coherent, adaptive navigation. Successful trajectories show GPT-6-Astra linking landmarks and actions into ordered routes. GPT-6-Astra grounds object descriptions and spatial relations in the current scene, uses successive observations to track completed and remaining instruction steps, and seeks additional views to resolve uncertain interpretations. By coordinating scene understanding with task progress, it can follow instructions through multiple turns and room transitions while adjusting its actions in response to new evidence. These behaviors demonstrate how general visual and linguistic capabilities can support sustained, autonomous navigation without task-specific training.

(3) Reliable route execution and goal verification remain challenging, even with ultra reasoning effort. Failure cases include substantial route deviations despite plausible local landmark matches, repeated difficulty traversing constrained passages, and arrival claims that conflict with the final position. Fifteen episodes fail under both reasoning settings. Of the 21 failures of GPT-6-Astra (ultra reasoning) in the 100-episode evaluation, 13 end at least 5 m from the reference goal, including seven more than 10 m away. These failures point to persistent difficulties in preserving route constraints, checking whether movement produces the intended progress, and confirming arrival at the instructed destination. Strong local scene understanding therefore does not consistently translate into correct route completion, even with ultra reasoning effort.

(4) The strong zero-shot capabilities of foundation models motivate us to rethink the role of embodied learning. GPT-6-Astra shows that a general-purpose model can combine instruction understanding, scene interpretation, and action selection to navigate effectively without navigation-specific fine-tuning. These results encourage future VLN research to build on the capabilities already present in foundation models. A key direction is to identify their remaining limitations and determine where navigation-specific methods and training can provide further improvements. Broader evaluations across diverse instructions, changing environments, and longer routes would help establish how well these capabilities generalize and guide the development of more capable and reliable navigation agents.

2. Evaluation Setup

2.1. Navigation Interface

Given a natural-language instruction, GPT-6-Astra navigates an unfamiliar indoor scene through low-level actions in continuous space [17]. Its visual input consists of 512×512 monocular RGB observations. Scene maps, reference trajectories, goal coordinates, global poses, and depth are unavailable through the navigation interface.

Each episode runs within a continuous Codex session. GPT-6-Astra itself interprets the instruction and available interaction context to select observations, action sequences, and when to stop. The Model Context Protocol (MCP) interface connects the model to the simulator (Figure 2). The task prompt specifies tool usage, budgets, and stopping conditions.

The `observe()` tool returns the current camera view without advancing the simulator or consuming an action step. The `step(actions)` tool executes an ordered se-

[1]The Codex harness maintains a continuous session in which GPT-6-Astra itself perceives, reasons, and makes navigation decisions.

[2]Following the common-used zero-shot evaluation protocol [10, 24, 25], we use the 100-episode R2R-CE subset introduced by Open-Nav [24]; trained-based results use the full validation-unseen split. Although not strictly fair, these comparisons help the VLN community assess the competitiveness and research value of zero-shot navigation with advanced foundation models.

Figure 2. **Navigation interface for GPT-6-Astra.** Within the Codex harness, GPT-6-Astra requests monocular RGB observations through `observe()` and executes primitive actions through `step(actions)`. The top-down scene and reference path are illustrative and are not provided to the model. The robot and camera sector are schematic; the path is a dataset reference, not an executed GPT-6-Astra trajectory.

quence of primitive actions: forward motion of 0.25 m, left/right turns of 15°, camera tilts of 30° up/down, and STOP. It returns execution counts, remaining budget, and termination status; obtaining a new RGB image requires another `observe()` call.

2.2. Evaluation Protocol and Metrics

Following prior commonon zero-shot VLN-CE evaluation protocols [24–26], we evaluate on the same 100-episode R2R-CE validation-unseen subset. This subset spans 10 scenes (see Footnote 2). We apply no navigation-specific fine-tuning and use neither a trained waypoint predictor nor a pre-built scene map.

We evaluate GPT-6-Astra with medium[3] and ultra reasoning effort, with one run per episode under each setting. Both configurations use the same task prompt and navigation tools. Each episode has a budget of 500 primitive actions and a 2,400 s time limit, ending upon STOP or exhaustion of either limit.

Success requires an explicit STOP with a final geodesic distance to the goal below 3 m. We report success rate (SR) and success weighted by path length (SPL) for task completion and path efficiency. Navigation error (NE), oracle success rate (OSR), and normalized dynamic time warping (nDTW) additionally measure final goal distance, whether any trajectory position enters the success radius, and reference-path agreement, respectively [15, 17]. Metrics are averaged over all 100 episodes in each setting.

[3]We explicitly use medium, the default reasoning effort for GPT-6-Astra in our Codex setup, and additionally evaluate ultra.

3. Navigation Performance

Table 1 summarizes navigation performance across learning regimes and observation settings. On the evaluated 100 episodes, GPT-6-Astra achieves 76.0% SR and 67.9% SPL with medium reasoning effort. Switching to ultra increases these scores to 79.0% and 69.5%, respectively, although the ultra run has slightly lower OSR and higher NE. Both configurations use monocular RGB observations and allow GPT-6-Astra to select observations, primitive actions, and when to stop, without navigation-specific fine-tuning.

Among zero-shot methods, the table covers both monocular navigation and approaches using panoramic observations or scene pre-exploration. GC-VLN reports 33.6% SR and 16.3% SPL with monocular observations, while SpatialAnt reports 66.0% SR and 54.4% SPL with panoramic observations and scene reconstruction [38, 43]. GPT-6-Astra achieves higher reported SR and SPL in both reasoning settings using monocular RGB alone, without a trained waypoint predictor or a pre-built scene map. Its strong completion rate is thus accompanied by strong success-weighted path efficiency.

Trained navigation methods provide a complementary reference. With monocular observations, Image2Nav reports 66.3% SR and 61.5% SPL [34]. With panoramic observations, Qwen-RobotNav-8B reaches 72.1% SR and 66.6% SPL [44]. Both GPT-6-Astra configurations record numerically higher SR and SPL, highlighting the competitiveness of a general-purpose model used without navigation-specific training. These trained-method results use the full validation-unseen split, while the table also includes subset evaluations. Differences in evaluation coverage and system settings make these comparisons contextual; they do not establish superiority under matched conditions.

Overall, these results demonstrate effective zero-shot navigation within the evaluated setting and motivate examining the behavior behind the scores. We next analyze how GPT-6-Astra connects instructions with successive observations and actions, and where route execution and goal verification remain unreliable.

4. From Instructions to Adaptive Navigation

We analyze GPT-6-Astra (ultra reasoning) on R2R-CE-100 to examine spatial grounding, route following, and action adjustment. Navigation metrics establish aggregate patterns, while recorded observations and actions explain individual behaviors.

Table 1. **R2R-CE val-unseen results.** Full: complete split; R2R-CE-100: Open-Nav's 100 episodes; Author-sampled: authors' samples with unverified Open-Nav correspondence; "–": unavailable or unverified. Pano.: 360° panoramic observations; Mono.: monocular views. NE: meters; other metrics: percentages. **Bold**/underline: best/second-best. Evaluation settings vary across methods.

Method	Source	Eval. split	View	NE ↓	OSR ↑	SR ↑	SPL ↑
VLN-CE supervised learning							
ScaleVLN [30]	ICCV'2023	Full	Pano.	4.8	–	55.0	51.0
ETPNav [2]	TPAMI'2025	Full	Pano.	4.7	65.0	57.0	49.0
BEVBert [1]	ICCV'2023	Full	Pano.	4.6	67.0	59.0	50.0
HNR [31]	CVPR'2024	Full	Pano.	4.4	67.0	61.0	51.0
Energy [20]	NeurIPS'2024	Full	Pano.	4.7	65.0	58.0	50.0
g3D-LF [29]	CVPR'2025	Full	Pano.	4.5	68.0	61.0	52.0
NavFoM [41]	ICLR'2026	Full	Pano.	4.6	72.1	61.7	55.3
ABot-N0 [9]	arXiv'2026	Full	Pano.	3.8	70.8	66.4	63.9
OmniNav [37]	ICLR'2026	Full	Pano.	3.7	74.6	69.5	66.1
Qwen-RobotNav-8B [44]	arXiv'2026	Full	Pano.	3.5	78.5	72.1	66.6
AstraNav-World [14]	arXiv'2025	Full	Pano.	3.9	73.9	67.9	65.4
NaVid [39]	RSS'2024	Full	Mono.	5.7	49.2	41.9	36.5
Uni-NaVid [40]	RSS'2025	Full	Mono.	5.6	53.3	47.0	42.7
NaVILA [8]	RSS'2025	Full	Mono.	5.2	62.5	54.0	49.0
Aux-Think [27]	NeurIPS'2025	Full	Mono.	5.9	54.9	49.7	41.7
Dynam3D [32]	NeurIPS'2025	Full	Mono.	5.3	62.1	52.9	45.7
StreamVLN [36]	ICRA'2026	Full	Mono.	5.0	64.2	56.9	51.9
DualVLN [35]	arXiv'2025	Full	Mono.	4.1	70.7	64.3	58.5
InternVLA-N1 [16]	Tech.Rep.'2025	Full	Mono.	4.8	63.3	58.2	54.0
D3D-VLP [33]	CVPR'2026	Full	Mono.	4.7	67.2	61.3	56.1
Image2Nav [34]	arXiv'2026	Full	Mono.	4.0	72.9	66.3	61.5
Qwen-RobotNav-8B [44]	arXiv'2026	Full	Mono.	4.4	72.7	65.7	59.6
Zero-shot VLN-CE							
NavGPT-CE-GPT4 [46]	AAAI'2024	Full	Pano.	8.4	26.9	16.3	10.2
HSGM [18]	CVPR'2026	Full	Pano.	5.4	58.7	47.9	32.8
MapGPT-CE-GPT4o [4]	ACL'2024	R2R-CE-100	Pano.	8.2	21.0	7.0	5.0
DiscussNav-GPT4 [21]	ICRA'2024	R2R-CE-100	Pano.	7.8	15.0	11.0	10.5
Open-Nav-GPT4 [24]	ICRA'2025	R2R-CE-100	Pano.	6.7	23.0	19.0	16.1
Three-Step Nav-GPT-5 [45]	AISTATS'2026	R2R-CE-100	Pano.	5.9	39.0	34.0	29.1
STRIDER-GPT-4o [13]	NeurIPS'2025	R2R-CE-100	Pano.	6.9	39.0	35.0	30.3
LaViRA-GPT-4o [11]	ICRA'2026	R2R-CE-100	Pano.	6.4	43.3	36.0	28.3
LaViRA-Gemini-2.5-pro [11]	ICRA'2026	R2R-CE-100	Pano.	6.5	48.7	38.3	28.3
O2C-Nav-Gemini-2.5-Pro [23]	arXiv'2026	R2R-CE-100	Pano.	5.8	65.0	49.3	30.9
EvoNav-GPT-4o [10]	CVPR'2026	R2R-CE-100	Pano.	6.0	35.0	30.0	24.9
EvoNav-Gemini-2.5-pro [10]	CVPR'2026	R2R-CE-100	Pano.	5.0	51.0	43.0	37.8
SmartWay-GPT-4o [25]	IROS'2025	R2R-CE-100	Pano.	7.0	51.0	29.0	22.5
SmartWay-GPT-5.5 [25]	IROS'2025	R2R-CE-100	Pano.	5.2	60.0	44.0	35.0
AgenticNav-Gemini-2.5-pro [19]	arXiv'2026	R2R-CE-100	Pano.	5.9	63.0	49.0	33.2
AgenticNav-GPT-5.5 [19]	arXiv'2026	R2R-CE-100	Pano.	5.2	65.0	55.0	48.4
SpatialNav [42]	arXiv'2026	Author-sampled	Pano.	5.2	66.0	64.0	51.1
SpatialAnt [43]	arXiv'2026	Author-sampled	Pano.	4.4	76.0	66.0	54.4
HarnessVLN-GPT-5.5 [7]	arXiv'2026	–	Pano.	4.0	72.7	60.8	43.5
Fast-SmartWay-GPT-4o [26]	arXiv'2025	R2R-CE-100	F3+P	7.7	–	27.8	25.0
CA-Nav [6]	TPAMI'2025	Full	Mono.	7.6	48.0	25.3	10.8
AO-Planner [5]	AAAI'2025	Full	Mono.	7.0	38.3	25.5	16.6
DreamNav [28]	arXiv'2025	–	Mono.	7.1	41.0	32.8	29.0
GC-VLN [38]	CoRL'2025	Full	Mono.	7.3	41.8	33.6	16.3
GPT-6-Astra (medium reasoning) (codex harness)	Report'2026	R2R-CE-100	Mono.	**2.7**	**83.0**	76.0	67.9
GPT-6-Astra (ultra reasoning) (codex harness)	Report'2026	R2R-CE-100	Mono.	3.0	82.0	**79.0**	**69.5**

Score sources. Source denotes the method's publication. NaVid, DualVLN, InternVLA-N1, NavFoM, ABot-N0, OmniNav, AstraNav-World, and Qwen-RobotNav scores follow Qwen-RobotNav Table 1 [44]. MapGPT, DiscussNav, and NavGPT CE scores follow SmartWay, Open-Nav, and CA-Nav, respectively [6, 24, 25]. LLM variants follow EvoNav Table 1, SmartWay Table I, and AgenticNav Table 1; SmartWay-GPT-5.5 is AgenticNav's reproduction [19]. LaViRA/Fast-SmartWay report three/four-run means; O2C-Nav uses main-table means. Each GPT-6-Astra row is a single run on the same 100 episodes.
Inputs and settings. F3+P: three frontal views with initial or on-demand 360° panoramas. O2C-Nav uses four surrounding views; STRIDER also uses Qwen-VL-Max for perception. SpatialAnt uses scene reconstruction and simulator depth.

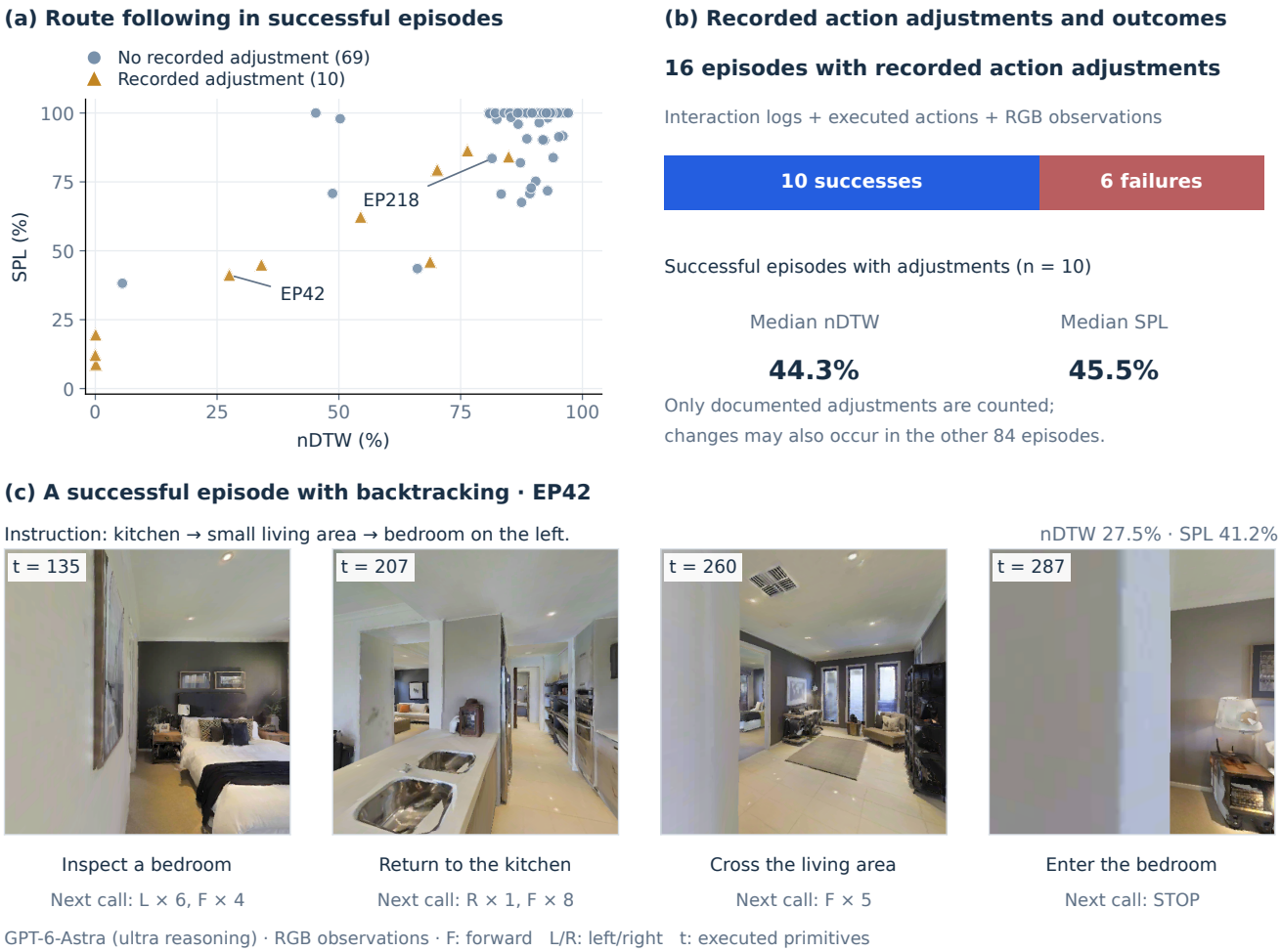

Figure 3. **Successful navigation can include substantial route deviations.** GPT-6-Astra (ultra reasoning) on R2R-CE-100: (a) nDTW and SPL for 79 successful episodes; (b) outcomes of 16 episodes with recorded action adjustments; (c) selected chronological observations of backtracking in EP42. Intermediate frames are omitted; action labels show only the next executed tool call.

Table 2. **Navigation performance of GPT-6-Astra (ultra reasoning) by instruction type.** Order-based selection uses cues such as *second* and *rightmost*; landmark-based selection uses *left/right of*, *between*, and *behind*. Scores (%) measure whole-task navigation performance.

Instruction subset	N	SR	SPL	nDTW
R2R-CE-100	100	79.0	69.5	73.3
Order-based selection	14	85.7	68.6	73.4
Landmark-based selection	13	61.5	59.6	71.8

4.1. How Does GPT-6-Astra Ground Spatial Instructions?

GPT-6-Astra can use target order and positions relative to landmarks to guide navigation. For this analysis, we distinguish two instruction subsets. *Order-based selection* specifies a target by its order among alternatives, such as the *second room* or *rightmost doorway*. *Landmark-based selection* specifies a target by its position relative to a reference landmark, using *left/right of*, *between*, or *behind*. Table 2 reports success on 12/14 tasks (85.7%) with order-based selection and 8/13 (61.5%) with landmark-based selection. These scores measure whole-task navigation performance, not the accuracy of individual spatial relations; scene layouts and route difficulty also differ across subsets.

EP218 illustrates order-based selection: after leaving the bedroom, GPT-6-Astra passes a bathroom identified in the interaction log as the first room on the left, then approaches the next doorway. EP244 illustrates landmark-based selection: GPT-6-Astra selects the doorway *to the left of the white double doors*. Both tasks succeed, with final NE of 2.6 m and 1.8 m, respectively.

4.2. Does Reaching the Goal Mean Following the Instructed Route?

Most successful episodes follow the reference path closely, although some reach the goal after substantial detours. Among the 79 successes, the median nDTW is 90.4%, while eight (10.1%) score below 50.0% (Figure 3a). SR alone therefore does not capture route following. Here, nDTW measures reference-path agreement rather than the correctness of individual instruction steps.

Specifically, in EP116, GPT-6-Astra moves from the closet through the bedroom to the bathroom and stops beside the tub and sink. The episode achieves an NE of 0.1 m and an nDTW of 89.2%, consistent with close reference-path agreement. In contrast, EP42 succeeds with an NE of 0.7 m, but its nDTW of 27.5% and SPL of 41.2% indicate a substantial detour.

4.3. How Does GPT-6-Astra Adjust Its Actions?

GPT-6-Astra can adjust its actions during navigation, but these adjustments do not guarantee successful or efficient completion. We identify 16 episodes with reported route deviations or passage difficulties, supported by changed actions and corresponding RGB observations. Ten episodes succeed and six fail (Figure 3b); the successful cases have median nDTW and SPL of 44.3% and 45.5%, respectively. These counts include only documented action adjustments and do not establish their causal effect on navigation success.

EP42 provides a concrete example of successful navigation with substantial backtracking (Figure 3c). After reaching a bedroom, GPT-6-Astra checks whether its route matches the instructed kitchen–living-room–bedroom sequence. It returns to the kitchen, follows the inner aisle, crosses the small living area, and enters the bedroom on the left. The recorded images and executed actions show this revisitation and change of route. Together, these results highlight the need to distinguish task completion, route following, and action adjustment when assessing navigation behavior.

5. Failure Modes and Reliability

On R2R-CE-100, GPT-6-Astra (ultra reasoning) succeeds in 79 episodes; three others enter the success radius but fail at termination, and 18 never enter it (Figure 4a). We examine route following, execution, and stopping, then compare failures across reasoning settings.

5.1. Where Does Route Following Break Down?

Many failed episodes end well beyond the success radius. Among the 21 failures of GPT-6-Astra (ultra reasoning), 13 end at least 5 m from the reference goal, including seven more than 10 m away. In EP513, the instruction specifies a fireplace living area, two white sofas, and the adjacent dining-room entrance. The final view contains a fireplace and pale seating, and GPT-6-Astra reports arrival, yet the final NE is 14.5 m. These local landmark matches do not verify arrival at the instructed destination. The recorded views do not locate the first route error.

5.2. Can GPT-6-Astra Recover from Execution Difficulties?

GPT-6-Astra can resume progress after execution difficulties, but recovery is not consistent. Five forward-only action batches across four evaluated episodes have pixel-identical RGB observations before and after execution. Three of these episodes eventually succeed. This indicator captures absent visible change; it does not confirm collisions or cover all execution difficulties.

In EP70, turns and forward movements near a narrow doorway are followed by views from inside the next room, and the episode succeeds. In EP176, eight forward actions produce identical RGB observations (Figure 4b). The episode later exhausts the 500-action budget with a final NE of 35.4 m. These outcomes distinguish temporary execution difficulties from eventual task failure: issuing actions alone does not establish movement or effective recovery.

5.3. Does GPT-6-Astra Stop at the Right Place?

Issuing STOP does not ensure arrival at the correct destination. GPT-6-Astra invokes STOP in 99 episodes; only EP176 reaches the action limit. Three failed episodes have OSR equal to one, indicating that their trajectories entered the success radius before termination. In EP705, GPT-6-Astra reports finding a fire extinguisher beside a doorway and stops with a final NE of 12.3 m, despite OSR equal to one. Recognizing an expected landmark and previously passing near the goal therefore do not ensure a correct final stop. OSR alone does not reveal when the trajectory entered the success radius or why it later ended elsewhere.

5.4. Does Higher Reasoning Effort Resolve These Failures?

Many navigation failures persist under both reasoning settings. Of the 24 tasks failed by GPT-6-Astra (medium reasoning), nine succeed with GPT-6-Astra (ultra reasoning), while 15 remain unsuccessful (62.5%). Conversely, six tasks succeed with medium reasoning but fail with ultra reasoning. Among the 15 shared failures, GPT-6-Astra (ultra reasoning) has lower final NE in eight tasks and higher NE in seven.

EP1133 illustrates partial improvement without task completion (Figure 4c). With both reasoning settings, GPT-6-Astra approaches the portrait, turns left, and stops facing the arched doorway. The final NE is 7.5 m with medium

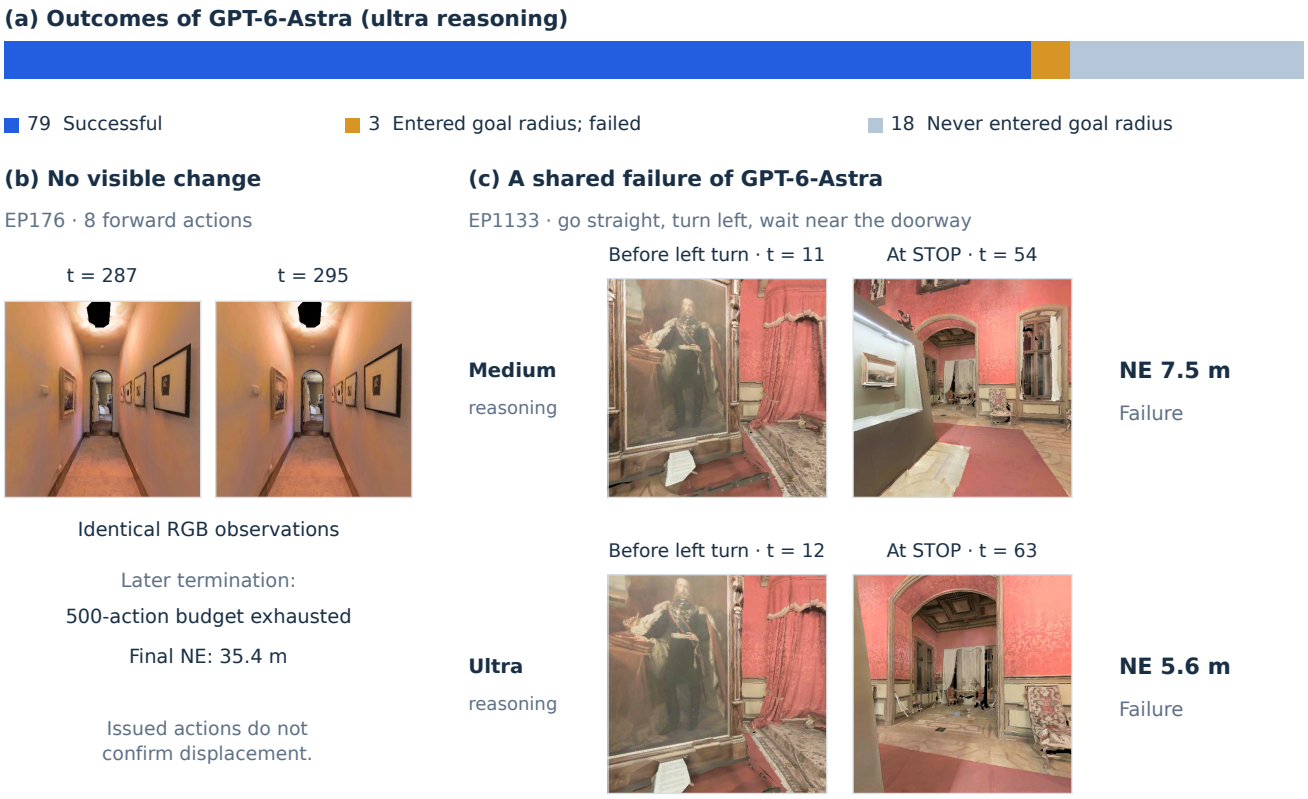

Figure 4. **Execution and goal-verification failures.** GPT-6-Astra (ultra reasoning) on R2R-CE-100: (a) navigation outcomes; (b) unchanged RGB observations after eight forward actions in EP176, which later exhausts the action budget. Identical images show no visible change, not a measured collision. (c) EP1133 fails under both reasoning settings despite a lower final NE with ultra reasoning. Frames show corresponding events at different action counts; t denotes executed primitive actions.

reasoning and 5.6 m with ultra reasoning; both exceed the 3 m success threshold.

6. Implications for Embodied Navigation

6.1. Do General-Purpose Models Offer Another Route to VLN?

GPT-6-Astra offers an alternative route to VLN by navigating directly through a minimal observation-and-action interface. Its reported success rate exceeds those of the trained navigation models in Table 1, without additional navigation-specific fine-tuning in our evaluation. This result suggests that competitive navigation need not require a separately trained navigation policy. General-purpose models can select views, execute movements, and decide when to stop through their existing capabilities. Comparisons under matched observations, action budgets, and task sets are needed to establish when this route is preferable to specialized training.

6.2. What Should Future VLN Methods Improve?

Future VLN methods should translate instruction and scene understanding into reliable route execution and goal verification. Our analysis identifies three priorities: preserving route constraints when landmarks recur, detecting ineffective motion and recovering, and verifying that the final location satisfies the instruction. These behaviors should be assessed alongside SR using reference-path agreement, execution feedback, and stopping outcomes. Navigation-specific training, memory, and action control should be evaluated by the measurable improvements they provide on these remaining failures. The shared failures across reasoning settings also show that higher reasoning effort alone does not ensure consistent task completion in the evaluated configurations.

6.3. How Should This Route Be Evaluated?

Broader, controlled evaluations are needed to establish the robustness of this route across navigation settings. The current findings concern R2R-CE-100 and the evaluated monocular interface. Matched comparisons should con-

trol observations, action spaces, budgets, and software versions. In the present runs, the Codex versions differ between medium and ultra reasoning (0.155.1 and 0.155.0), so their paired outcomes do not isolate the effect of reasoning effort. Recorded observations and actions support analysis of observable behavior, but do not establish the accuracy of individual semantic decisions or reveal internal reasoning. Future evaluations should combine endpoint success with route fidelity, recovery, and correct stopping, and test whether these capabilities persist across unfamiliar environments and longer instructions.

7. Conclusion

We evaluated GPT-6-Astra on R2R-CE-100 through a minimal interface for monocular RGB observations and primitive actions within the Codex harness. GPT-6-Astra achieves 76.0% SR with medium reasoning and 79.0% with ultra reasoning, without navigation-specific fine-tuning in our evaluation. The analysis shows spatial grounding, ordered room transitions, and route revision, alongside persistent difficulties in execution and goal verification. Fifteen tasks fail under both reasoning settings. These results identify an alternative route to embodied navigation through general-purpose models and motivate further work on reliable route following and task completion.

Limitations and future work. Our evaluation is limited to R2R-CE-100, with one run per episode under each reasoning setting. Future evaluations will report means and standard deviations over repeated runs and cover additional datasets, including RxR-CE and NavRAG-CE. High inference cost and latency currently constrain practical deployment. Historical reductions in inference cost at comparable capability levels [22] suggest that these barriers may diminish as models and inference systems improve. Finally, GPT-6-Astra is a proprietary model whose training data are not available for inspection. We therefore cannot determine whether navigation data were used in training or, if so, their amount and composition. Zero-shot in this report means that we apply no navigation-specific fine-tuning; it does not establish that the model (by OpenAI) has never encountered navigation data.

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